

CSC1042 Computer Network II



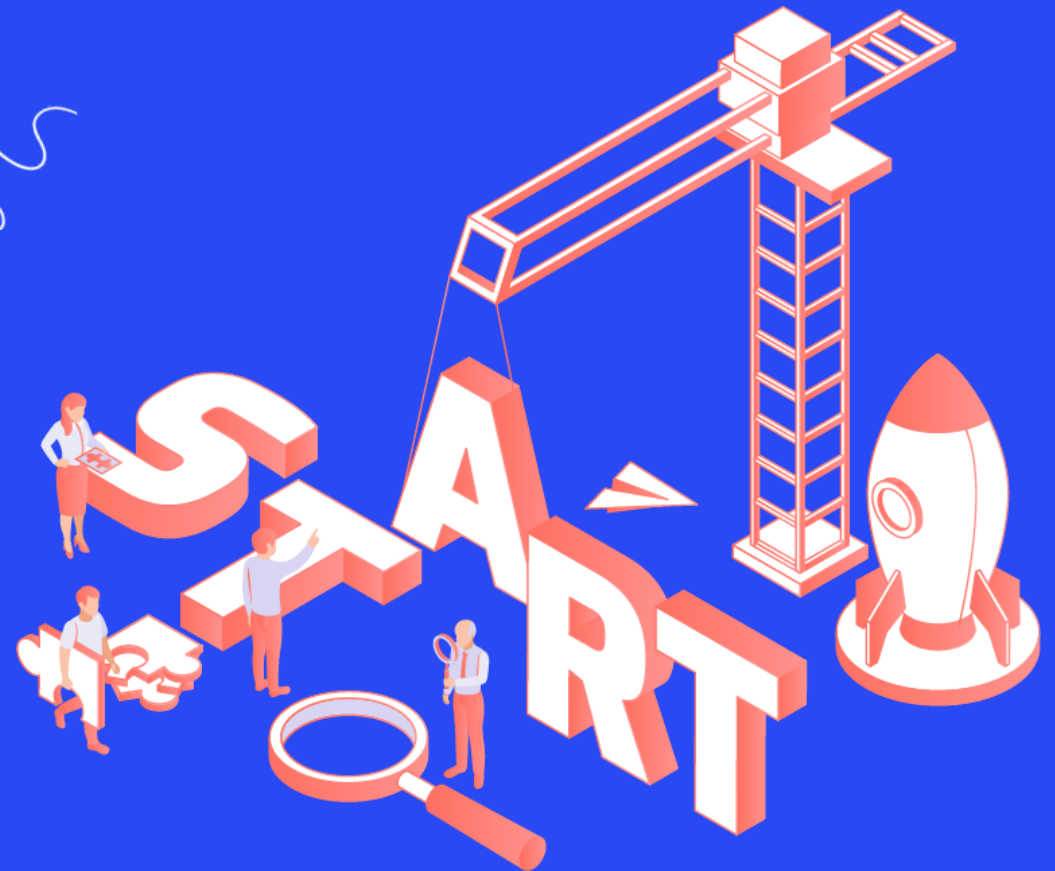
CSC1010 Crash Course

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Learning outcomes

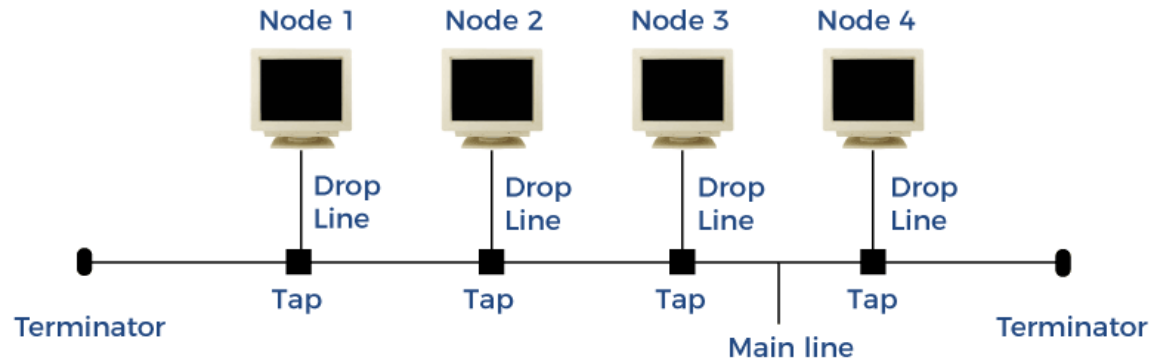
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- Define what is a network along with network types and network topologies
 - Describe transmission media
 - Describe network devices used to build various networks
 - Enumerate the layers of the OSI and TCP/OP models and explain what each layer does
 - Describe the concept of data encapsulation
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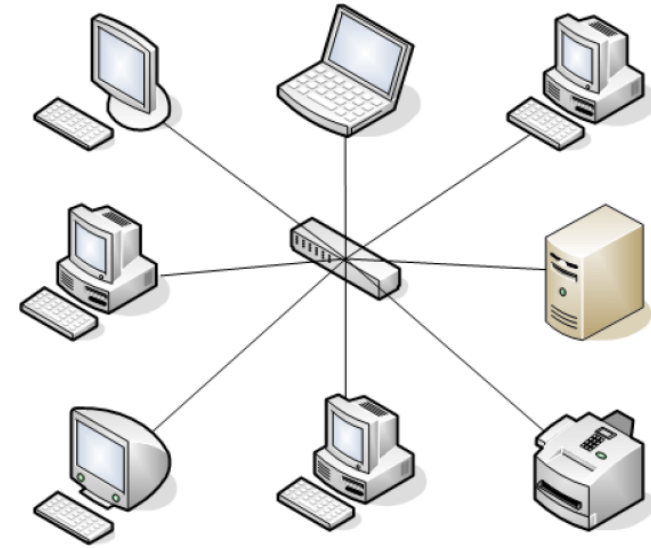
Recap

- Two or more devices linked together to share resources, e.g., files, printers, processing power, storage capacity, etc.
- There are different types of networks
 - Personal Area Network (**PAN**): e.g., smartwatch connected to a phone
 - Local Area Network (**LAN**): e.g., home or office
 - Metropolitan Area Network (**MAN**): e.g., company's offices that are far apart within the same city
 - Wider Area Network (**WAN**): e.g., bank branches across a country

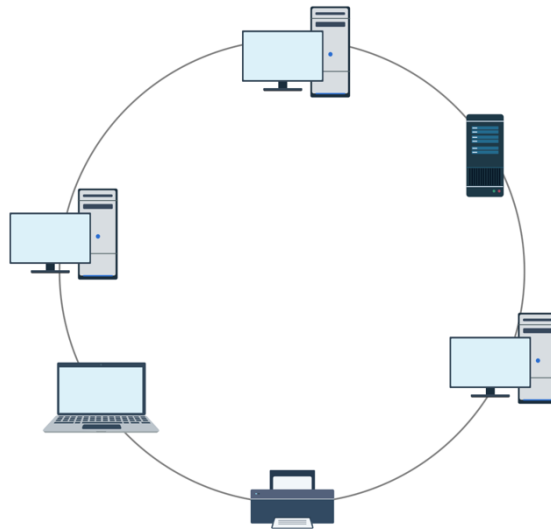
Network topologies



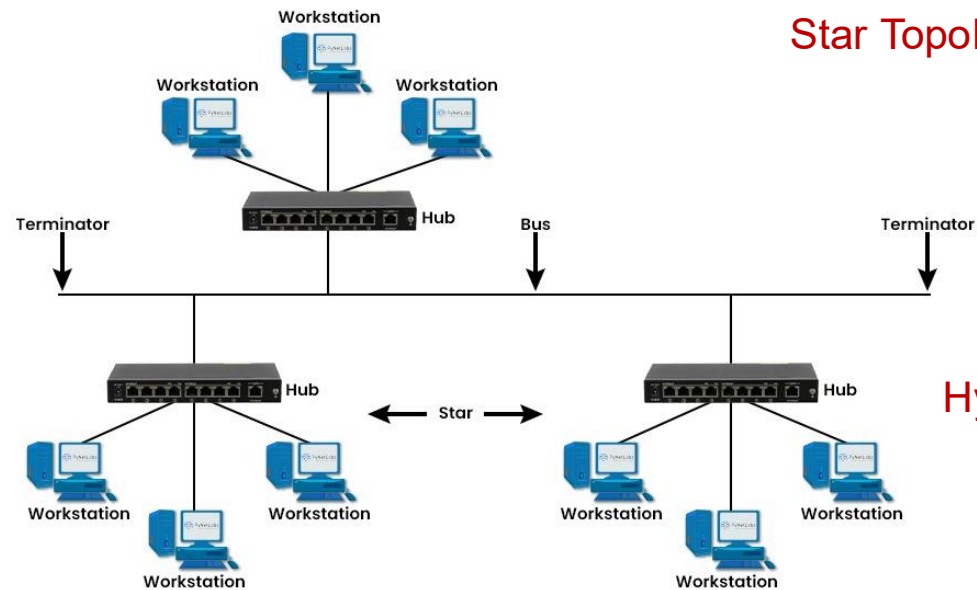
Bus Topology



Star Topology

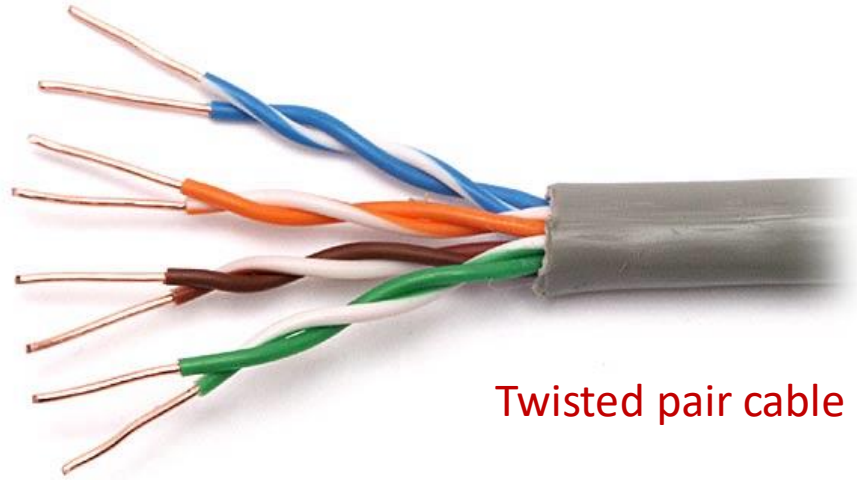


Ring Topology



Hybrid Topology

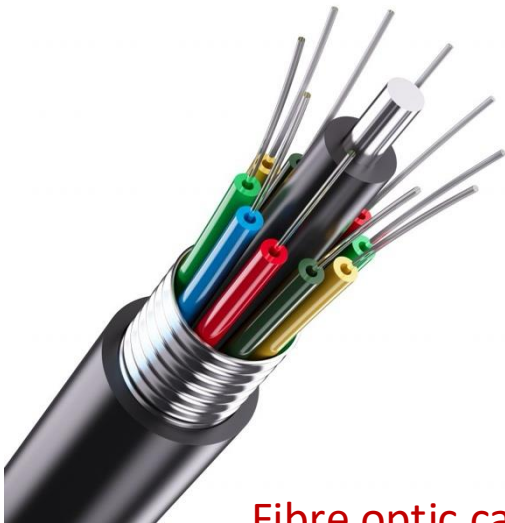
Transmission media



Twisted pair cable



Coaxial cable



Fibre optic cable



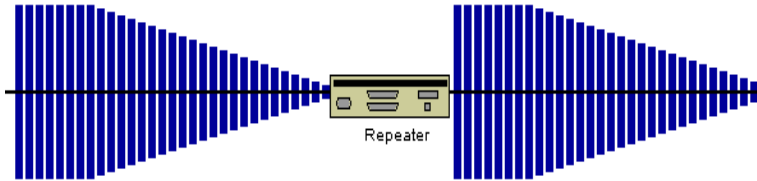
Radio waves

Network devices



Network Interface Card (NIC)

- used to connect devices to networks
- Each NIC has a unique ID, called MAC address



Repeater

- Operates at the physical layer
- Amplifies and regenerates the signal
- Extends the distance at which the signal can be transmitted

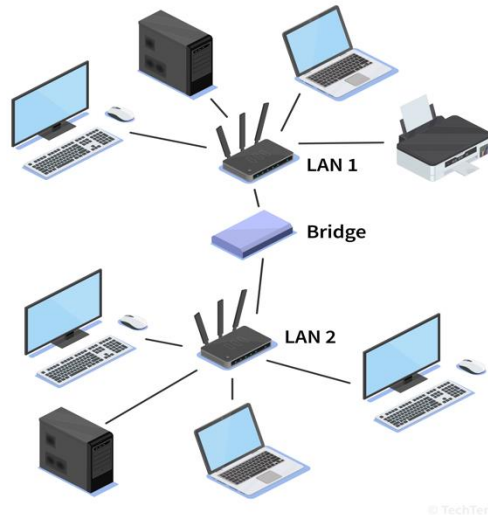
Hub



Hub

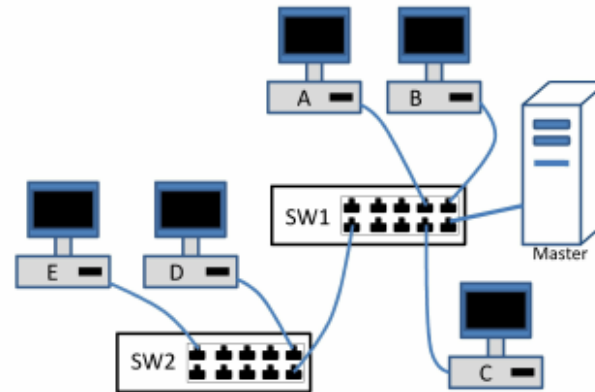
- Operates at the physical layer
- It's basically a multi-port repeater
- Data received from one port will be sent to all connected devices
- It has no intelligence

Network devices



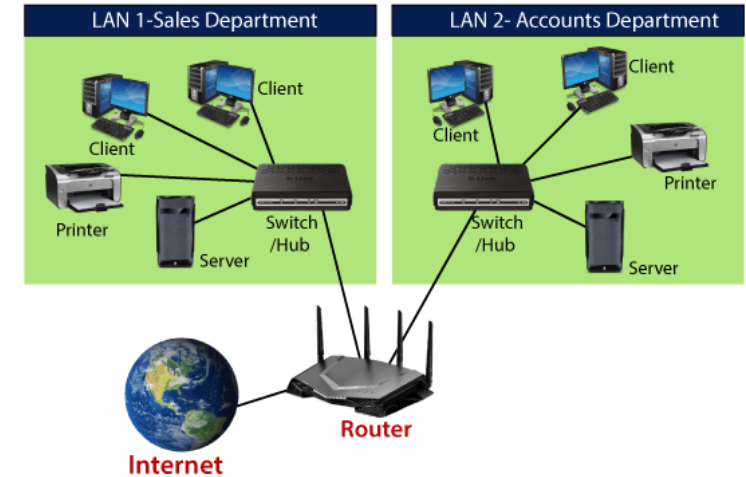
Bridge

- Operates at the Data Link layer
- Similar to a repeater but perform data filtering using MAC addresses
- Connects multiple LANs that use the same protocol to create a larger LAN



Switch

- Operates at the Data Link layer
- Can connect multiple LANs
- Filter data based on MAC address
- Can perform error checking before forwarding data
- Some switches operate at Network layer. They are called Layer 3 switches

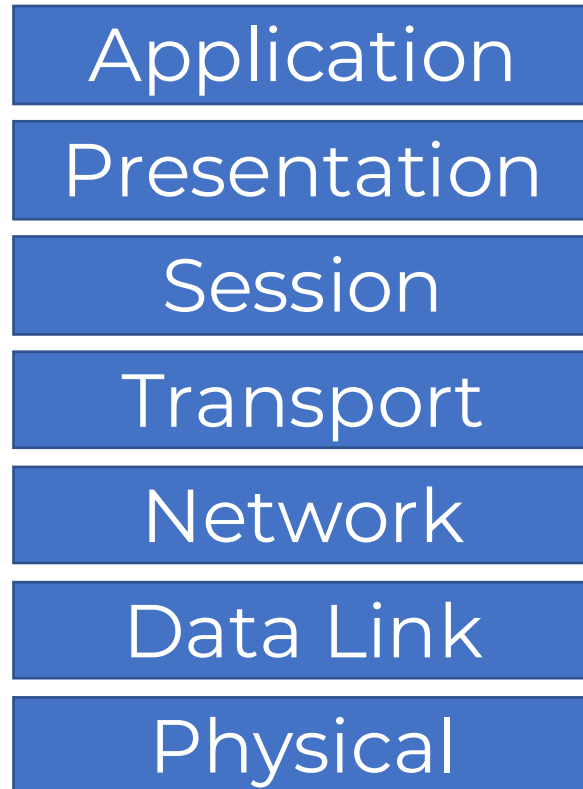


Router

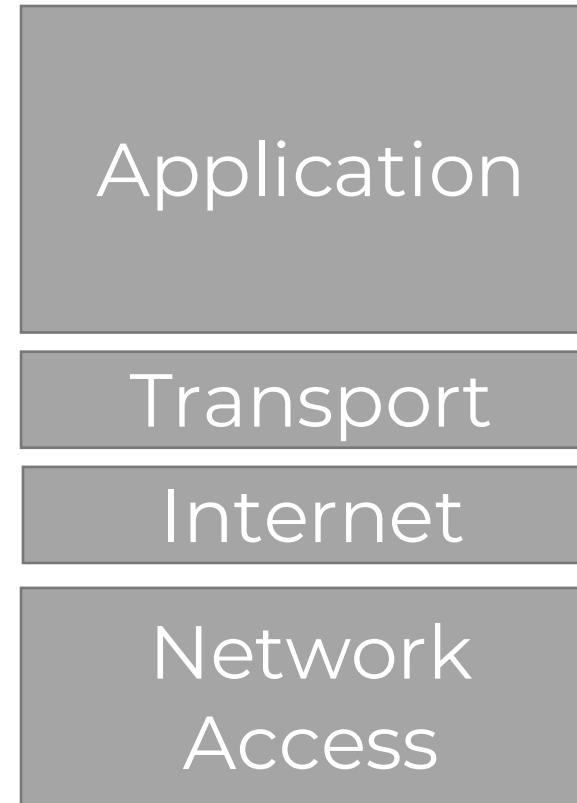
- Operate at the Network layer
- Route data based on IP addresses
- Connect LANs and WANs
- Maintain routing tables, used to route data from source to destination
- Can also perform error checking before forwarding data

Network models

OSI

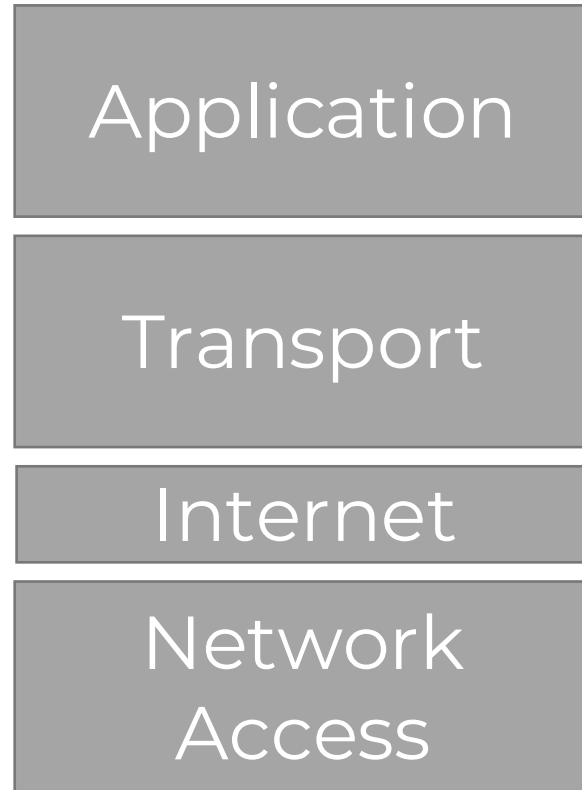


TCP/IP

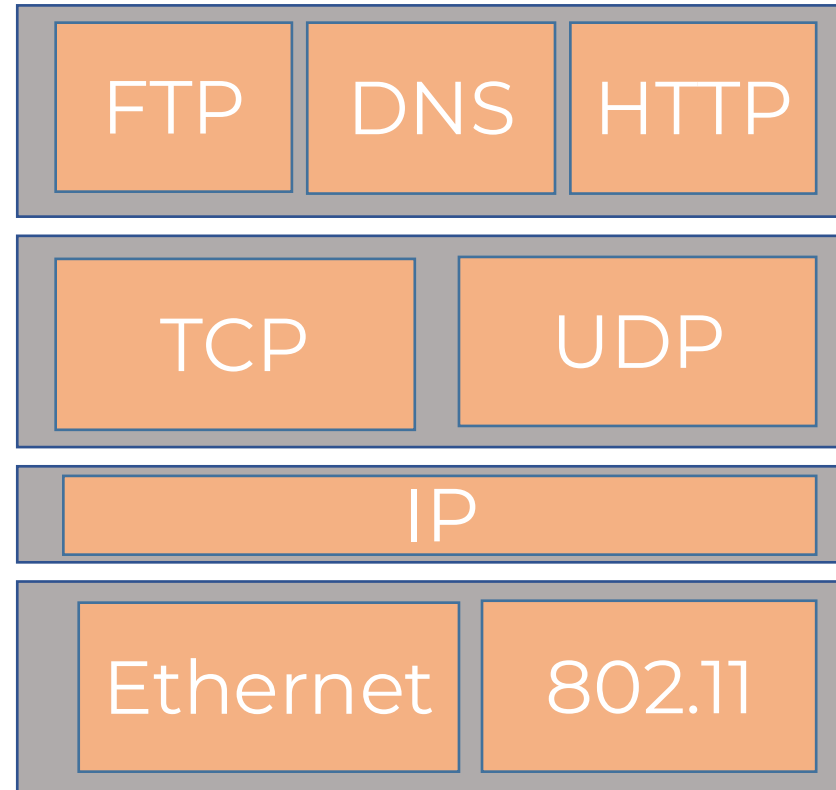


Network models

TCP/IP

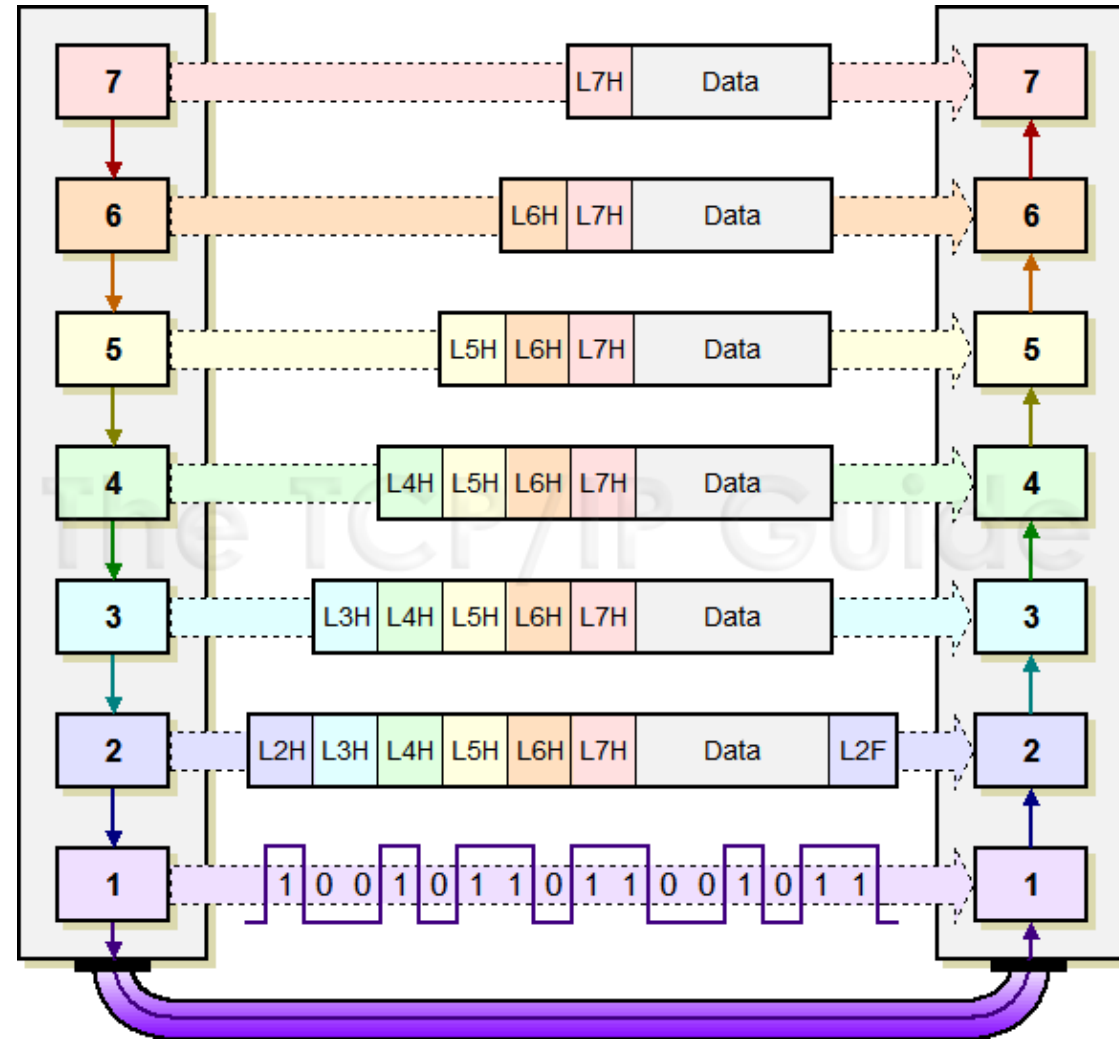


Protocol Suite

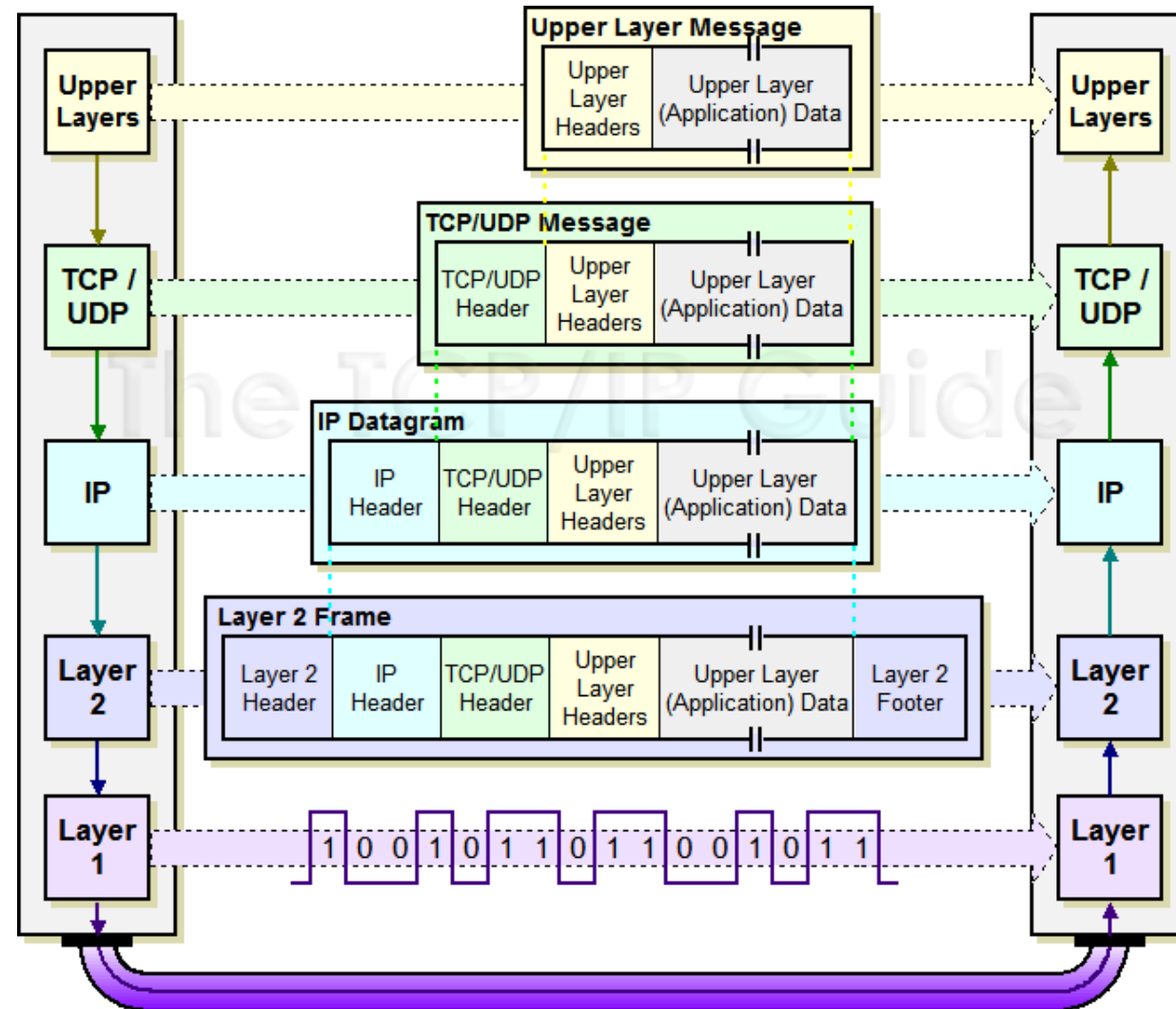


Data encapsulation (OSI model)

- Each protocol creates a *protocol data unit (PDU)* that includes headers required by that protocol and data to be transmitted



Data encapsulation (TCP/IP model)



THANK YOU